

E-Commerce Sales Dashboard

Project Objective

Analyze e-commerce sales data to identify trends in revenue, customer purchasing behavior, and product performance. The dashboard supports business decision-making through interactive KPIs and visualizations.

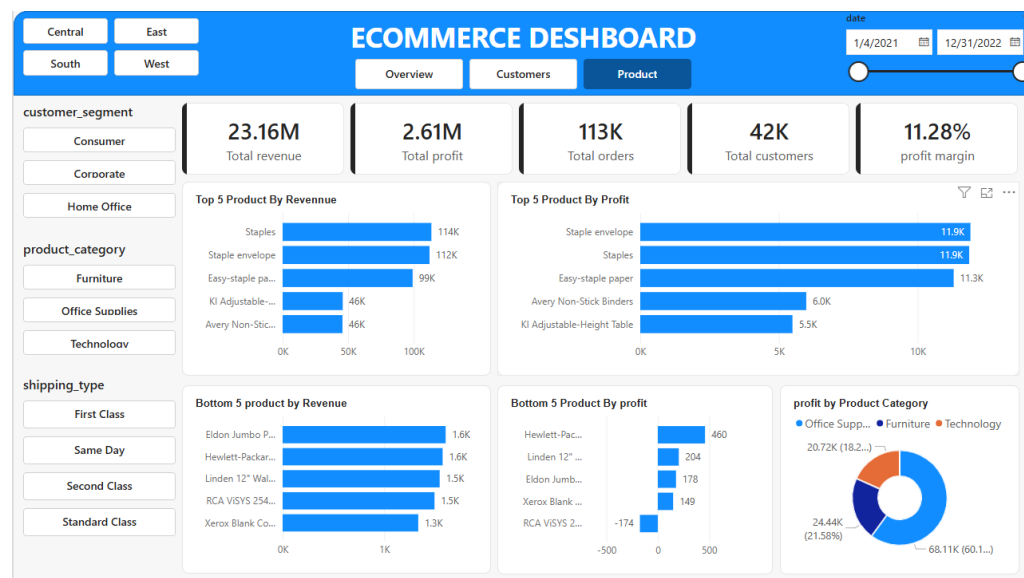
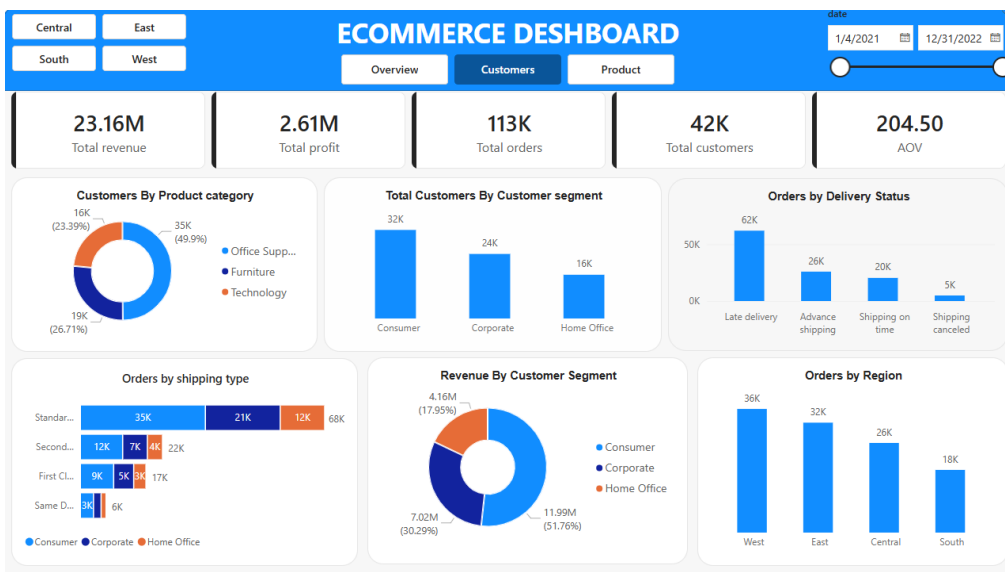
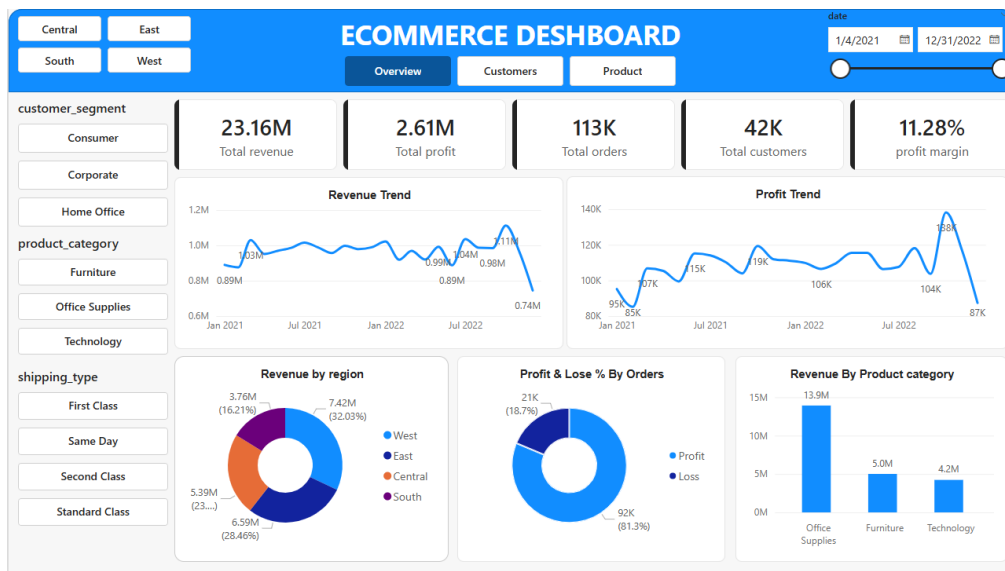
Tools Used

- Microsoft Power BI
- Power Query (M Language)
- DAX

Attribute	Details
Dataset	E-Commerce Sales Dataset
Records	113,000+
Columns	21
Time Period	April 2021 – April 2023

Dashboard Pages

- Executive Overview
- Customer Analysis
- Product Analysis



Data Preparation

Data Cleaning & Transformation

Data Preparation Steps

- ✓ Imported dataset into Power BI
- ✓ Verified data types
- ✓ Removed duplicate records
- ✓ Created calculated columns
- ✓ Built DAX measures
- ✓ Validated data quality
- ✓ Created business KPIs

Data Quality Validation

During data validation, the following checks were performed:

- Checked for missing values
- Verified revenue and profit consistency
- Validated delivery dates
- Identified invalid delivery records

Data Quality Issue

Approximately **17.7%** of delivery-related records contained inconsistent delivery dates.

To ensure accurate reporting, delivery metrics were excluded from the final dashboard.

DAX Measures Created

- Total Revenue
- Total Profit
- Total Orders
- Total Customers
- Profit Margin
- Average Order Value

Business Insights

Executive Summary

This analysis identified key trends in revenue, customer behavior, and product performance. The insights below summarize the most important findings that can support business decision-making.

1. Sales Performance

Observation:

- **Total Revenue: \$23.16M**
- **Total Profit: \$2.61M**
- **Profit Margin: 11.28%**

Business Insight:

The business generated strong overall revenue while maintaining a healthy profit margin. This indicates that revenue growth is also contributing positively to profitability rather than being driven solely by higher sales volume.

2. Customer Insights

Observation:

- The **Consumer** segment generated the highest revenue.
- Consumer customers accounted for the largest share of total orders.

Business Insight:

Consumer customers represent the primary customer base. Retaining this segment through loyalty programs and personalized marketing could have the greatest impact on future revenue growth.

3. Product Performance

Observation:

- Office Supplies generated the highest category revenue.
- The Top 5 products contributed a significant portion of total revenue.

Business Insight:

Revenue is concentrated among a relatively small number of high-performing products. Maintaining inventory availability and promotional support for these products could help sustain sales performance.

4. Customer Purchasing Behavior

Observation:

- Standard Class was the most frequently selected shipping method.
- Customers purchased products across multiple categories, with Office Supplies being the most common.

Business Insight:

Customers appear to favor standard shipping, which may indicate a preference for lower shipping costs over faster delivery. The popularity of Office Supplies suggests consistent demand within this category.

Data Quality Note

Observation:

- Delivery-related fields contained approximately 17.7% invalid records.

Business Insight:

To maintain the accuracy of the analysis, delivery performance metrics were excluded from the final dashboard until the underlying data quality issues can be resolved.

Business Recommendations

Product Strategy

Focus promotional campaigns on high-performing product categories while reviewing the pricing and positioning of low-performing products.

Customer Strategy

Develop customer retention programs for the Consumer segment to maintain revenue growth.

Sales Strategy

Analyze underperforming regions to identify opportunities for improving customer acquisition and sales performance.

Dashboard Improvements

Future enhancements may include:

- Customer Lifetime Value (CLV)
- Customer Retention Analysis
- Year-over-Year Sales Growth
- Forecasting
- Delivery Performance Analysis after resolving data quality issues